Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

# SDC Hackathon

# Technical Tools Declaration

**Hosted by Xiamen University Malaysia (XMUM) AI Club**

**Submission Deadline: 1 November 2026**

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

***Instagram:*** @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

# 1. TEAM AND PROJECT OVERVIEW

## Team Name

T010NG

## Project / Solution Name

Kotak

## Name (Leader)

Hee Zi Jie

## Email & Phone (Leader)

heezijie2004@gmail.com, +60125118986

## University Affiliation(s)

Asia Pacific University of Technology & Innovation (APU), University of Reading and University of Sheffield

## Live Prototype URL

https://kotak-alaskxntxna.replit.app (mirror: https://kotak.tp073654.workers.dev)

## Code Repository URL (public / access granted to judges)

https://github.com/TolongLabs/Kotak

# 2. ARCHITECTURE & TECHNOLOGY STACK

## 2.1. System Architecture Overview

*Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.*

***[Maximum 200 words]***

A user opens Kotak in English, Malay or Chinese. The React app loads dated JSON snapshots built from O*NET, the Anthropic Economic Index and DOSM Job Market Insights, and computes every verdict, fit score and roadmap in the browser with fixed rules. A three.js scene stands all 923 jobs by fit. Answers stay in the browser's **localStorage**.

The app calls a Hono API on the same origin. The API matches typed job titles and writes cited explanations with Claude Haiku 5.5, then checks each against the rules, falling back to a template. It also judges uploaded proof of a finished roadmap step with Claude's vision, and keeps only the verdict. Kopi's lines go to ElevenLabs for speech; both are cached by request hash. Ask Kopi and the MCP endpoint call 8 read-only tools over the same rules; two read Neon Postgres totals for skill gaps and job ads.

A RunPod GPU built each job's 6 skills and 18 questions with Qwen3.8-27B, and translated ten jobs into Malay and Chinese; both ship inside the snapshots.

GitHub Actions deploys the app and API to Cloudflare Workers on every merge to **main**, and refreshes data weekly. Replit publishes the same code. Keys live only in GitHub, Cloudflare and Replit secrets.

System block diagram: see Appendix A.

# 2. ARCHITECTURE & TECHNOLOGY STACK

## 2.2. Languages, Frameworks & Libraries

*List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.*

| Name & Version | Purpose in Project | License |
|---|---|---|
| TypeScript 7.0 | The language of the whole codebase | Apache 2.0 |
| Bun 1.4 | Runtime, package manager, test runner and Replit server | MIT |
| React 19.3 | The web app's user interface | MIT |
| Vite 8.3 | Builds and serves the web app | MIT |
| Tailwind CSS 4.3 | Styling and design tokens | MIT |
| Recharts 3.10 | The Future Skills Radar and the roadmap charts | MIT |
| Lucide React 1.61 | Icons | ISC |
| pixi.js 6.5 and pixi-live2d-display 0.4 | Renders Kopi, the Live2D cat | MIT |
| Hono 4.13 | The API server on Cloudflare Workers and Bun | MIT |
| Zod 4.6 | Validates every request, response and data file | MIT |
| Anthropic TypeScript SDK 1.1 | Calls the Claude API | Apache 2.0 |

# 2. ARCHITECTURE & TECHNOLOGY STACK

## 2.2. Languages, Frameworks & Libraries (continued)

| Name & Version | Purpose in Project | License |
|---|---|---|
| MCP server SDK 2.3 | Serves Kotak's tools to AI assistants at **/api/mcp** | Apache 2.0 |
| Neon serverless driver 1.2 | Connects the API to Postgres | MIT |
| Wrangler 4.151 | Deploys to Cloudflare Workers | MIT or Apache 2.0 |
| Playwright Core 1.64 and Clearcote 0.60 | Drives the browser for the JobStreet scraper | Apache 2.0, BSD-3 |
| three 0.186, @react-three/fiber 9.8 and postprocessing 6.39 | Draw the Two Futures 3D space and its bloom | MIT, MIT, Zlib |
| fflate 0.8 and PGlite 0.5 | Unpack the source archives; Postgres for the API's tests | MIT, Apache 2.0 |
| Live2D Cubism Core for Web | Runtime that plays Live2D models, loaded from Live2D's CDN | Live2D Proprietary Software License |
| Atkinson Hyperlegible and Shantell Sans fonts | Body, mono and Kopi's handwritten type, from Google Fonts | SIL Open Font License 1.1 |
| Playwright 1.64 | Films the pitch video's demo takes from a local production build | Apache 2.0 |
| Remotion 4.0.530 | Assembles and renders the pitch video's live-demo segment | Remotion License, free for individuals |
| ffmpeg 6.1 | Encodes the demo footage and sets its loudness | LGPL 2.1 or later |
| Adobe Premiere Pro | Edits the pitch video: the slides, the narration (slide parts at 1.2× with pitch kept) and the demo segment | Adobe Creative Cloud subscription |

# 3. APIs & CLOUD SERVICES

## 3.1. External APIs

*List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.*

| API and Provider | Purpose in Project | Access Tier (free / paid / sponsor-provided) |
|---|---|---|
| Claude API, Anthropic | Matches job titles, explains verdicts, runs Ask Kopi and judges step proof | Paid |
| Text to Speech API, ElevenLabs | Kopi's voice | Paid |
| RunPod Serverless API, RunPod | The two GPU batches: each job's skills and questions, then ten jobs' translations | Sponsor-provided credit |
| Google Fonts, Google | Serves the app's fonts | Free |
| Live2D Cubism CDN, Live2D | Serves the Cubism Core runtime for Kopi | Free |

The data sources in section 4.2 are downloaded files, not live APIs: the app never calls them at runtime. API keys are not shown in this document.

# 3. APIs & CLOUD SERVICES

## 3.2. Cloud Platforms & Deployment

*List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.*

| Service & Provider | Purpose in Project | Plan / Region |
|---|---|---|
| Replit Deployments, Replit | Publishes the live demo from **main** on the team's own account | Replit Core (sponsor-provided) |
| Cloudflare Workers, Cloudflare | The mirror deployment: app, API, KV cache and a Durable Object for MCP and chat | Workers Free; global, Durable Object in APAC |
| Neon Postgres, Neon | The national view's sample rows, JobStreet listings, the admins' reviews and matched skills, and admin accounts | Free; **aws-ap-southeast-1**, Postgres 18 |
| RunPod Serverless, RunPod | The vLLM GPU endpoint for the two build-time batches; 0 workers when idle | Sponsor credit; one H100 80 GB GPU |
| GitHub and GitHub Actions, GitHub | Code hosting, CI, deploys on every merge to **main** and the Saturday data refresh | Free |

# 4. AI TOOLS, MODELS & DATASETS

## 4.1. AI Models Used in the Solution

*List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.*

| Model & Version | Source / Provider | Usage (pre-trained / fine-tuned / self-trained) | License / Terms |
|---|---|---|---|
| Claude Haiku 5.5 (**claude-haiku-5-5**) | Anthropic, Claude API | Pre-trained | Anthropic Commercial Terms of Service |
| ElevenLabs **eleven_v4** and **eleven_v4_turbo**, voice Johari | ElevenLabs | Pre-trained | ElevenLabs Terms of Service |
| Qwen3.8-27B-FP8 | Qwen, run on RunPod with vLLM | Pre-trained | Apache 2.0 |

No model is fine-tuned or self-trained. Qwen3.8-27B-FP8 ran twice before the demo, both times on RunPod: once to write the skills and questions of 923 jobs, and once to translate ten of those jobs into Malay and Simplified Chinese. Claude Haiku 5.5 matched 2,030 SkillsFuture Singapore roles to O*NET and linked their skills to 269 jobs' skill areas in 592 requests (about US$0.56); a validator kept every code inside O*NET and each job's own matched roles. None of these runs happens at run time.

# 4. AI TOOLS, MODELS & DATASETS

## 4.2. Datasets

*List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.*

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| O*NET 30.1 Database | https://www.onetcenter.org/database.html | Public | CC BY 4.0 | N |
| Anthropic Economic Index, April and May 2026 | https://huggingface.co/datasets/Anthropic/EconomicIndex | Public | Data CC BY; code MIT | N |
| DOSM Job Market Insights, Q3 2019 to Q2 2026 | https://www.dosm.gov.my | Public | DOSM terms, with source credit | N |
| OpenDOSM skill-related underemployment by age | https://open.dosm.gov.my/data-catalogue/lfs_qtr_sru_age | Public | CC BY 4.0; one figure cited | N |
| UN SDG indicator 4.4.1, Malaysia | unstats.un.org/sdgapi, indicator 4.4.1 | Public | UN terms of use; one baseline cited | N |
| SkillsFuture Singapore Skills Framework 2.0 | jobsandskills.swda.gov.sg | Public | SWDA terms; verbal approval, 31 Oct 2026 | N |
| Kotak skills and questions for 923 jobs | Built from O*NET by the RunPod batch | Synthetic, from public data | The team's own work | N |

# 4. AI TOOLS, MODELS & DATASETS

## 4.2. Datasets (continued)

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| Question explainers and reversed question wording | Drafted from the Kotak questions by Gemini (explainers) and DeepSeek (reversed wording); a script checked every line and the team spot-checked samples | Synthetic, from public data | The team's own work | N |
| JobStreet job listings | Collected by the team's scraper from https://my.jobstreet.com, reviewed by admins | Self-collected | JobStreet's terms apply; Kotak publishes totals only | N; job ads only |
| CIP 2020 to SOC 2018 Crosswalk | nces.ed.gov/ipeds/cipcode | Public | Public domain (US government work) | N |
| Malaysian regulated professions | Each row cites its Act on the Laws of Malaysia site, the regulator's site or a legal database | Self-collected | The team's own work; sources cited | N |

# 4. AI TOOLS, MODELS & DATASETS

## 4.2. Datasets (continued)

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| Malay and Chinese translations | App text, SDG text and 3 demo jobs with Claude; 10 more jobs by Qwen3.8-27B on RunPod, validated | Synthetic, from public data | The team's own work | N |
| What-would-you-do questions for 902 jobs | Written from the Kotak skills and O*NET tasks by Claude Sonnet and GPT fast, script-checked, samples reviewed | Synthetic, from public data | The team's own work | N |
| SDG tags and reasons | Judged against the UN target text by Claude Haiku 5.5, sampled and rechecked by Claude Sonnet 5.5 | Synthetic, from public data | The team's own work | N |
| ESCO to O*NET-SOC Crosswalk | onetcenter.org/crosswalks/esco | Public | CC BY 4.0; ESCO by the EC | N |
| Demo profiles | Written by the team | Synthetic | The team's own work | N |

MyCOL and Epoch AI are cited as context only; Kotak loads no data from them. Approved JobStreet listings feed the job-title suggestions, the Job Ads in Malaysia card, the role cards' ad counts and pay, and Kopi's **job_market** tool, but no verdict, fit score or roadmap.

# 4. AI TOOLS, MODELS & DATASETS

## 4.3. AI-Assisted Development Tools

*Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.*

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

| Tool & Version | Used For | Extent of Use (minor / moderate / major) |
|---|---|---|
| Claude Code, with Claude models | Code, reviews, research and documentation | Major |
| Codex CLI, OpenCode and Antigravity CLI, with GPT, DeepSeek, GLM and Gemini models | Parallel code changes from written briefs | Moderate |
| Claude, GPT and Gemini chat models | The field simulation and design review in **docs/research** | Minor |
| OpenAI Whisper, **whisper-1** API | Word timings for the pitch video's cuts and captions | Minor |

# 4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

***[Maximum 100 words]***

Every change ran type-checks and a build in CI, and a team member reviewed it before it reached the main branch. The team checked each feature by running the app in a browser at phone and desktop widths, not by trusting the tool's report. Facts in the app and in these documents link a dated public source. AI-drafted skills carry a visible label in the app until a person reviews them. The team owns every merged line and corrected tool mistakes before merging.

# 5. SPONSOR-PROVIDED RESOURCES

*Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.*

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

| Sponsor | Resource (API / credits / platform / data) | How It Was Used in the Project |
|---|---|---|
| Replit | Platform, Core | Hosts the live demo, published from **main** on the team's own account |
| RunPod | Credits | A serverless H100 ran two batches, about US$3.04 of credit in all: the skills and questions of 923 jobs (783 seconds, about US$1.25), then the Malay and Chinese translations of ten jobs (174 requests, 276,636 prompt and 82,652 completion tokens, about US$1.79) |
| SDC mentors (KrackedDevs) | Mentoring | One mentor session on 23 October 2026 |

# 5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

***[Maximum 100 words]***

RunPod was essential to build Kotak's data: it turned O*NET's tasks into 6 skills and 18 questions for each of 923 jobs, which no person could write by the deadline, and it translated ten of them into Malay and Chinese. Both batches run once, so the live app does not depend on RunPod, and the Claude Message Batches API could replace it. Replit hosts the submitted live demo; the same build also runs on Cloudflare Workers, so the hosting is replaceable.

# 6. SECURITY, PRIVACY & RESPONSIBLE USE

## 6.1. Data Handling & Privacy

*What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.*

***[Maximum 150 words]***

Kotak has no user accounts. A check-up runs in guest mode: the job, goal and answers stay in the browser's **localStorage**, and Settings > Reset All clears them.

The app sends a typed job title to the Claude API for matching, and a role's skill data for explanations. Ask Kopi sends the user's message, and the results of tools run on their check-up, to Claude. Kopi's lines go to ElevenLabs for speech. An uploaded proof file goes to Claude once to be judged and is never stored or logged; only the verdict stays in the browser. Neither request carries a name, email or ID. The API caches answers and audio by a hash of the request.

Postgres holds the national view as a role code, 6 skill levels, a sample flag and a date, with totals only for groups of 10 or more; no screen sends check-ups, so it holds labelled sample rows. The JobStreet scraper stores public job ads, not personal profiles. Keys live only in host secrets.

# 6. SECURITY, PRIVACY & RESPONSIBLE USE

## 6.2. Known Limitations & Risks

*Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.*

***[Maximum 150 words]***

- **Hallucination**: Claude only explains verdicts that fixed rules set. A check rejects any explanation that changes a verdict or lacks a dated citation, and a template replaces it.
- **Wrong job match**: the user confirms every AI title match.
- **Unreviewed skills and translations**: an open model drafted each job's skills and the Malay and Chinese text. Validators reject tasks from other jobs and translations that drop a term, and the app labels AI-drafted skills.
- **Thin trend data**: the Economic Index has two monthly points, so 12-month projections are rough. Every target is labelled an estimate, and thin data shows Unknown.
- **Data bias**: O*NET describes US jobs, and DOSM counts only online job ads.
- **Single providers and rate limits**: cached answers, templates and text cover Claude or ElevenLabs outages and HTTP 429 or 529 errors. A limit of 30 Claude calls a minute protects the key.
- **Proof judge**: it fails closed on any error and cannot confirm the certificate is the user's own, so the team says so in the Q&A.

# 7. COMPLIANCE & READINESS

*Tick all boxes that apply before submitting.*

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

# 8. DECLARATION

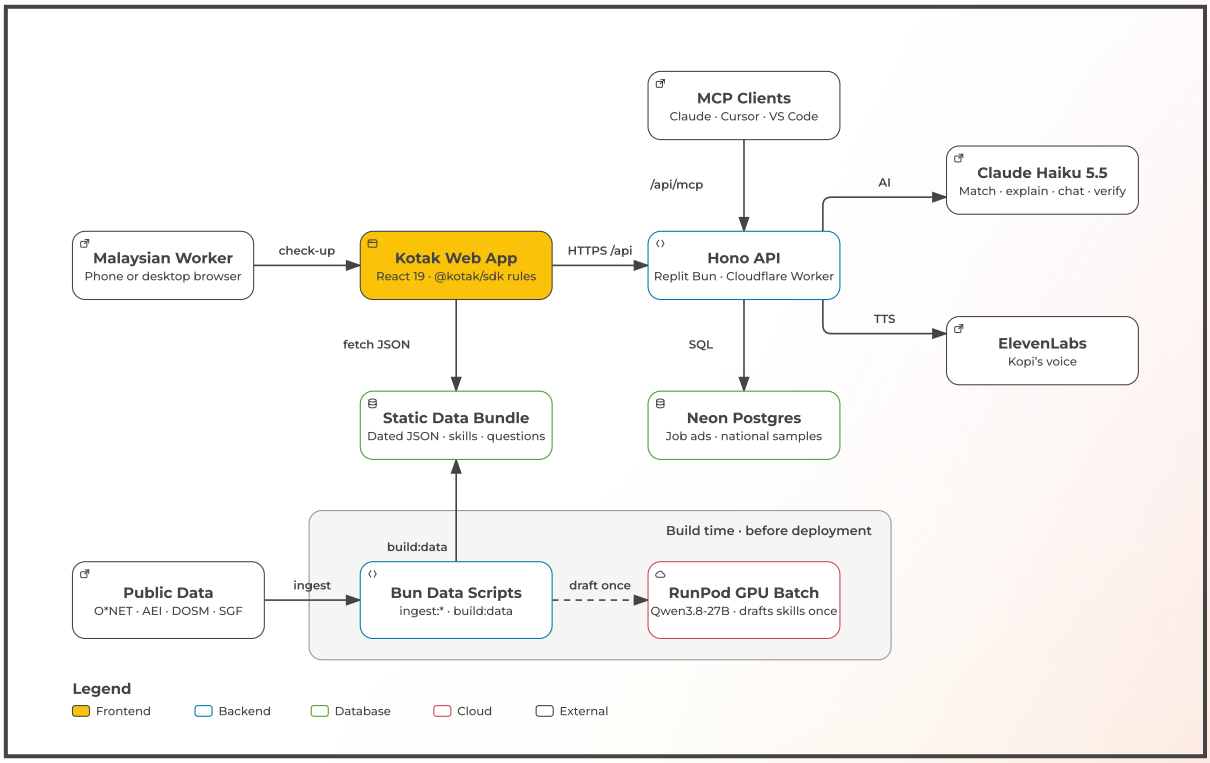
By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



31 October 2026

# 2. ARCHITECTURE & TECHNOLOGY STACK

## A. Appendix A: System Block Diagram




Dated public data and RunPod skill drafts feed the static bundle before deployment. The browser computes every verdict; the Hono API connects Claude, ElevenLabs, Neon Postgres and MCP clients. Generated with Archify.

# 5. SPONSOR-PROVIDED RESOURCES

## B. Appendix B: Replit Proof of Use

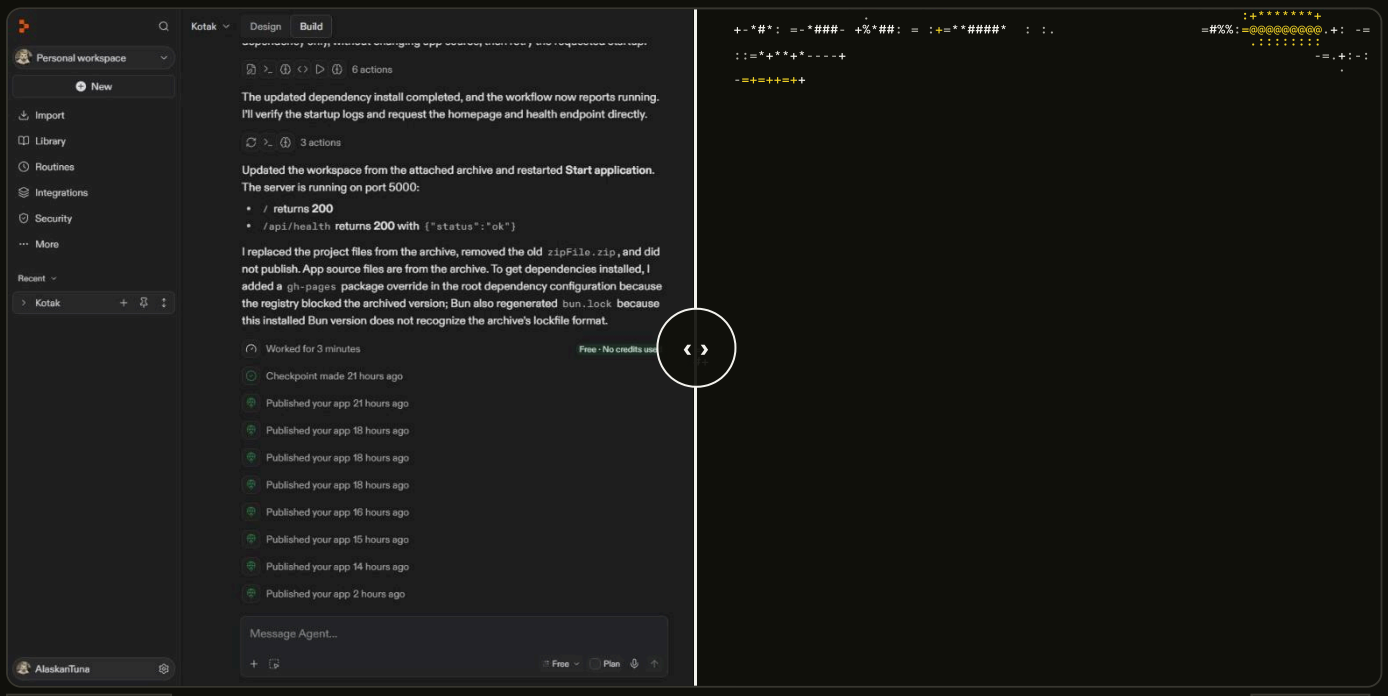
Handbook 10.6 asks for proof that Replit was used, showing the team's own account and project. The following pages show the live deployment at https://kotak-alaskxntxna.replit.app and the project https://replit.com/@alaskxntxna/Kotak with its publish and Git history, from the team leader's account. The same file is **submission/SDC2026_T010NG_Replit_POU.pdf** in the repository.

Proof of Use 23 Oct 2026 Status **Live**, public

# Replit

Kotak is **live on Replit Autoscale**, published from the team leader's own account and built from the team's GitHub repository.

- kotak-alaskxntxna.replit.app
- @alaskxntxna/Kotak
- Autoscale · Asia




WORKSPACE Fig 1, the Kotak project — the same screen, redrawn in glyphs GLYPHS


01 How Kotak uses Replit p. 2
02 Handbook checklist p. 2
03 How it runs p. 2
04 Replit figures 1–7 p. 3
05 Sources p. 6


**Live**
public deployment, Replit Autoscale

**200**
from /api/health, checked 23 Oct 2026

**2 vCPU**
4 GiB RAM, region Asia

**3**
secrets in Replit, values hidden

01

# How Kotak uses Replit

Source: the Replit workspace, 23 Oct 2026

| | | |
|---|---|---|
| 01 | **Replit account** | alaskxntxna (display name AlaskanTuna), the team leader's own account |
| 02 | **Project** | https://replit.com/@alaskxntxna/Kotak |
| 03 | **Live deployment** | https://kotak-alaskxntxna.replit.app, status Live, sharing Public |
| 04 | **Deployment type** | Replit Autoscale, region Asia, 2 vCPU / 4 GiB RAM |
| 05 | **Source** | GitHub repository TolongLabs/Kotak (named sdc-hack when captured), connected through Replit Git (branch `github-main`) |
| 06 | **Run configuration** | `.replit`: `bun install`, `bun run build`, then `bun run start` on port 5000 |
| 07 | **Secrets** | `ANTHROPIC_API_KEY`, `ELEVENLABS_API_KEY` and `SESSION_SECRET` in Replit Secrets; values stay hidden |
| 08 | **Publish history** | 9 publishes by AlaskanTuna between 22 and 23 October 2026 |

02

# Handbook checklist

Handbook 10.6 asks for any one proof; this document gives two

**[x] A live deployment Replit URL** — Fig 2 1 2 3 · Fig 7

`kotak-alaskxntxna.replit.app` is Live and Public, and serves Kotak.

**[x] A Replit project link with its history** — Fig 1 2 4 · Fig 3 1 · Fig 5 2

`replit.com/@alaskxntxna/Kotak`, with its publish and Git history.

**[x] The team's own account** — Fig 1 1 · Fig 2 1 · Fig 4 3

AlaskanTuna, the team leader, is signed in and publishes the app; commits go through the linked GitHub profile.

**[x] The team's own project** — Fig 4 1 2

The project's Git remote is the team's repository, TolongLabs/Kotak (then sdc-hack), through an active GitHub connection.

03

# How it runs

Config: .replit in the repository root

.replit — deployment

```
[deployment]
deploymentTarget = "autoscale"
build = ["sh", "-c", "bun install && bun run build"]
run = ["sh", "-c", "PORT=5000 bun run start"]
```

health.sh — checked 23 Oct 2026

```
curl https://kotak-alaskxntxna.replit.app/api/health

# response
{"status":"ok"}
```

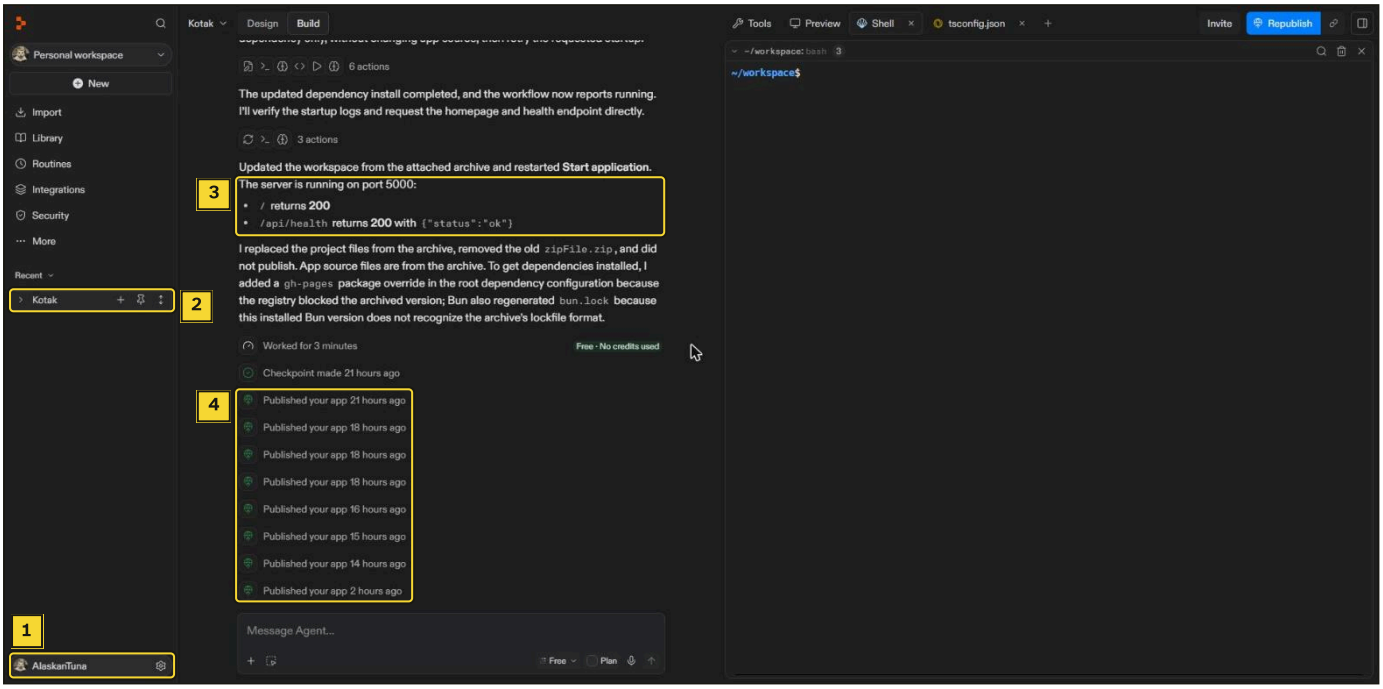
04

# Replit figures

Captured from the team leader's account, 23 Oct 2026

## FIG 1 The Kotak project

Build view

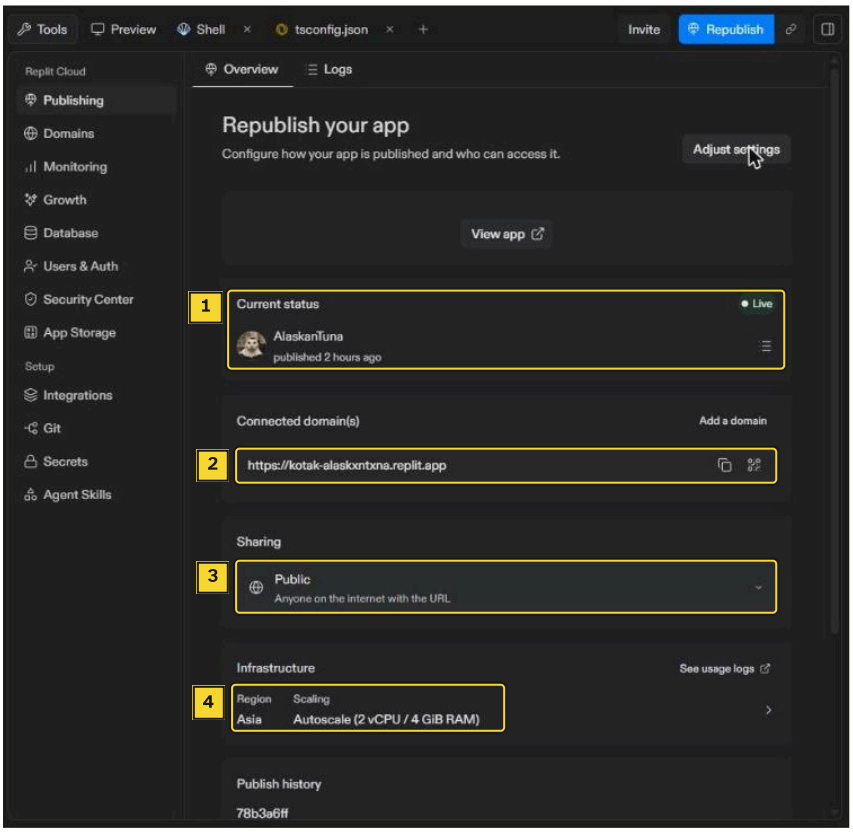



1 Signed in as AlaskanTuna, the team leader.

2 The Kotak project in the personal workspace.

3 Replit Agent's check: the server runs on port `5000` and `/api/health` returns `200`.

4 Publishes, from 21 hours to 2 hours before capture.




## FIG 2

## Publishing

Overview tab

1 Current status Live, published by AlaskanTuna.

2 The deployment URL, kotak-alaskxntxna.replit.app.

3 Sharing Public: anyone with the URL can open it.

4 Region Asia, Autoscale with 2 vCPU and 4 GiB RAM.

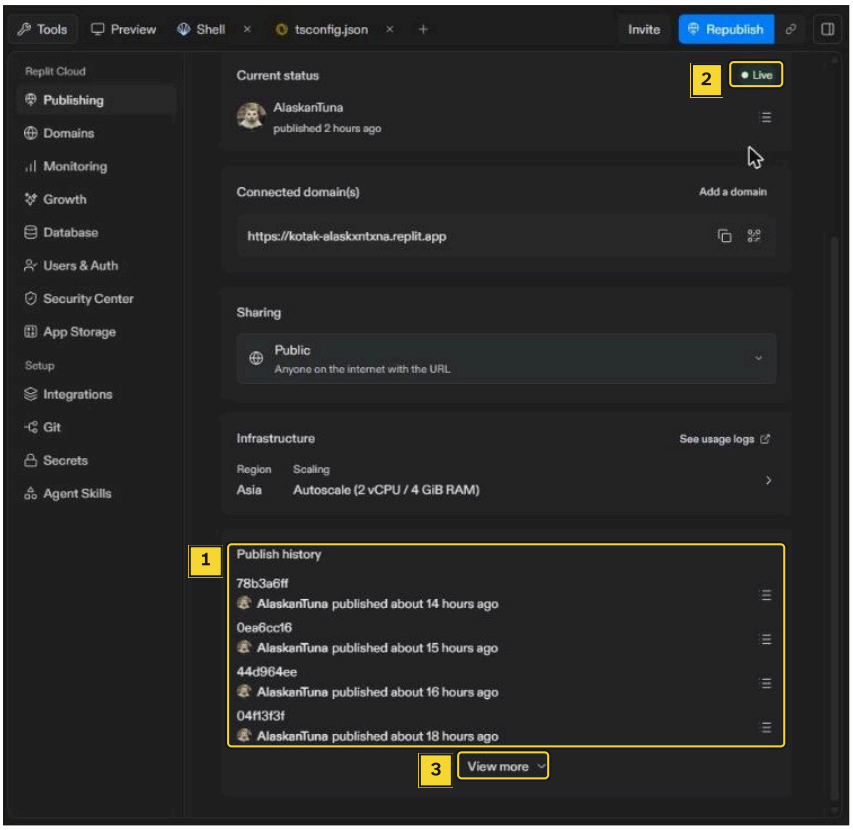



**FIG 3**

## Publish history

`Publishing, scrolled down`

1. Each publish carries its commit (78b3a6ff, 0ea6cc16, 44d964ee, 04f13f3f) and AlaskanTuna.
2. The deployment is still Live.
3. Older publishes sit behind View more.

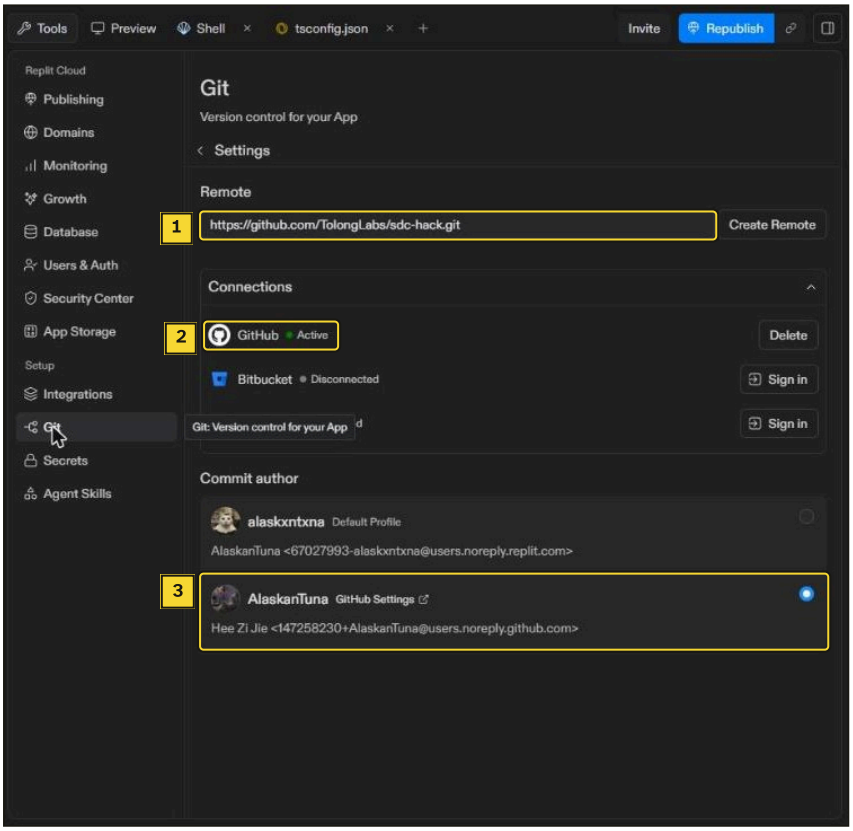



**FIG 4**

## Git settings

`Git, Settings`

1. The remote is the team's repository, github.com/TolongLabs/sdc-hack.git.
2. The GitHub connection is Active.
3. The selected commit author is the linked GitHub profile, AlaskanTuna.

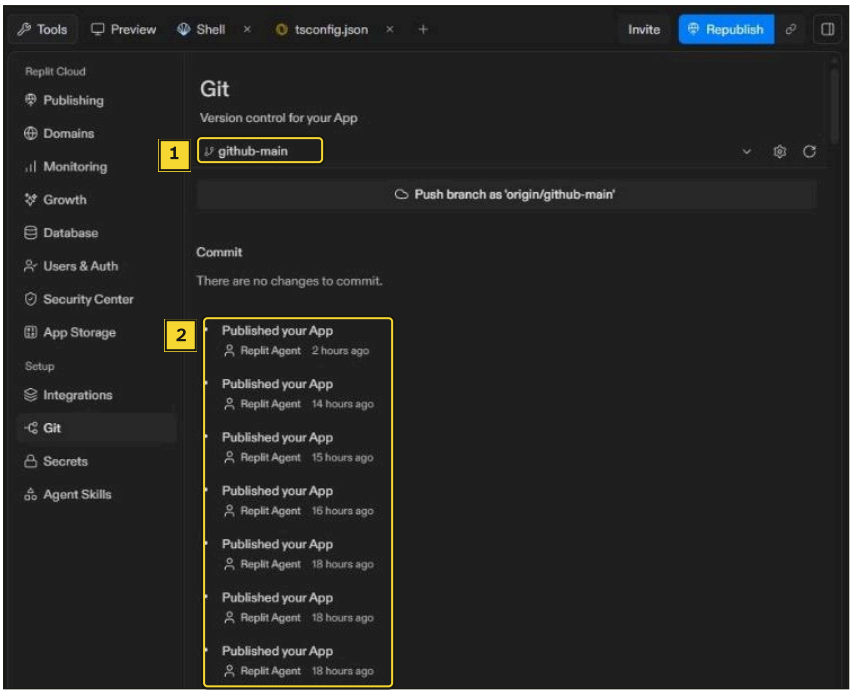



FIG 5

## Git history

`branch github-main`

1 Branch github-main, tracking the team's repository.

2 Publishes by Replit Agent, the newest 2 hours before capture.

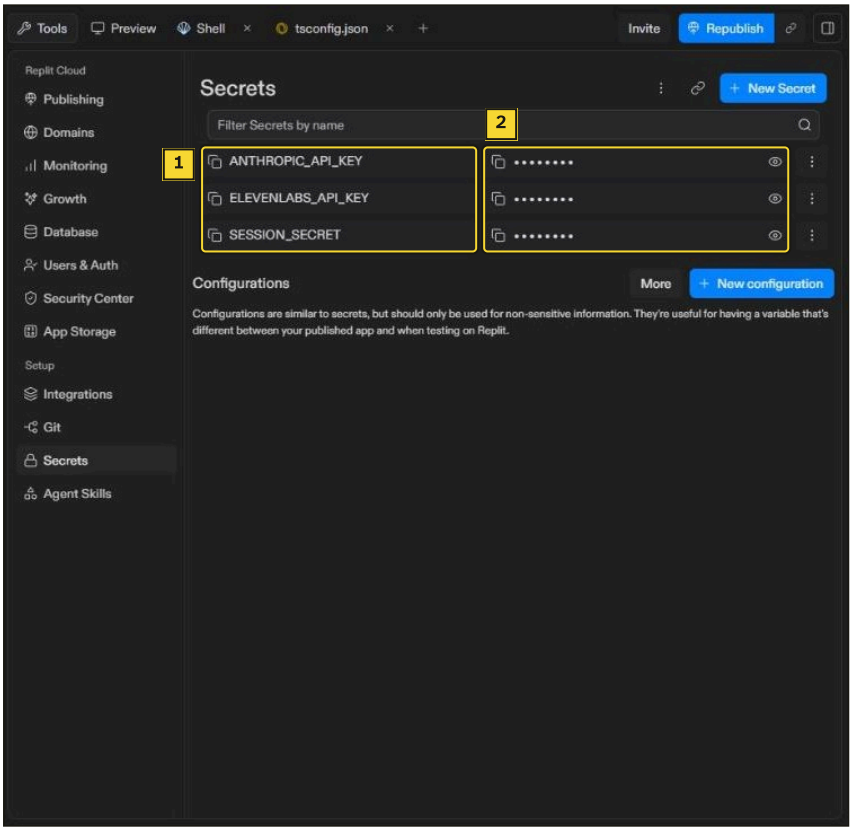



FIG 6

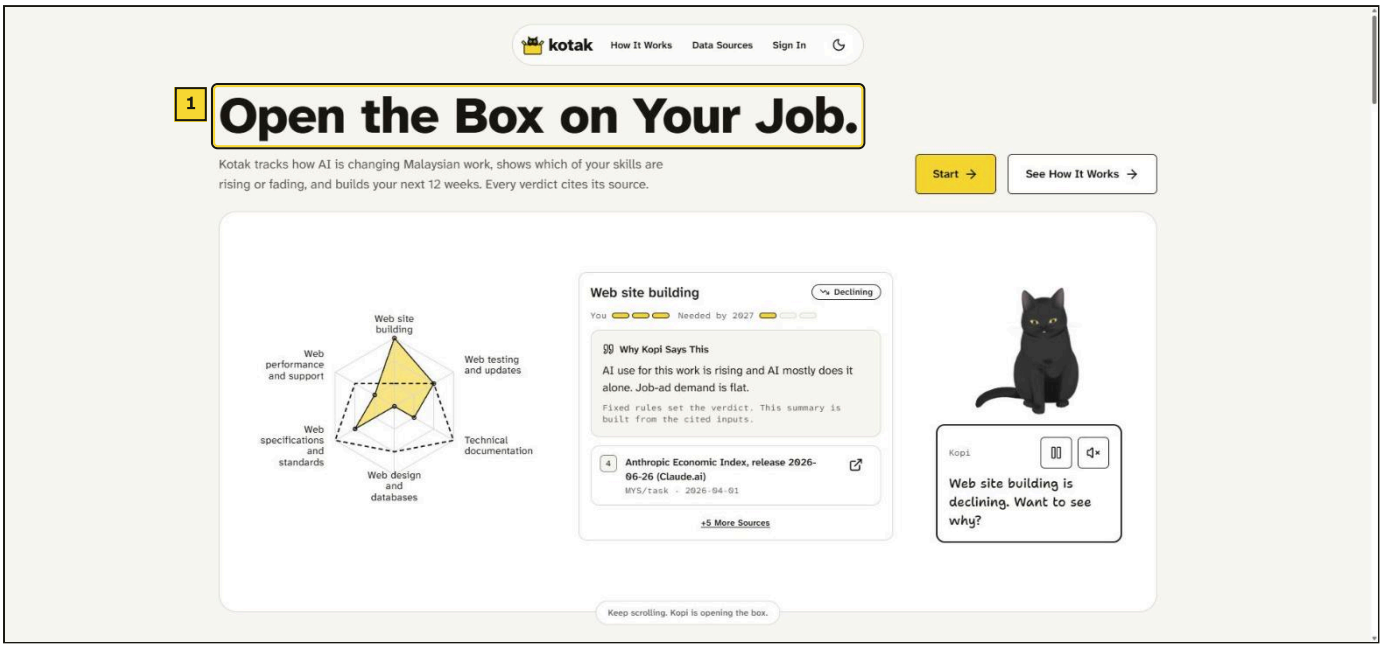
## Secrets

`Secrets tab`

1 ANTHROPIC_API_KEY, ELEVENLABS_API_KEY and SESSION_SECRET.

2 Values stay hidden, and none are in the code.

**FIG 7** **The published app**  23 Oct 2026




1 Kotak's landing page, opened from the deployment in Fig 2 on 23 Oct 2026.

05 · Sources

# Open it yourself: Kotak is live on Replit.

| | |
|---|---|
| Live app | kotak-alaskxntxna.replit.app |
| Project | replit.com/@alaskxntxna/Kotak |
| This PDF | scripts/proof-of-use/replit.html |
| Repository | github.com/TolongLabs/Kotak |





# 5. SPONSOR-PROVIDED RESOURCES

## C. Appendix C: RunPod Proof of Use

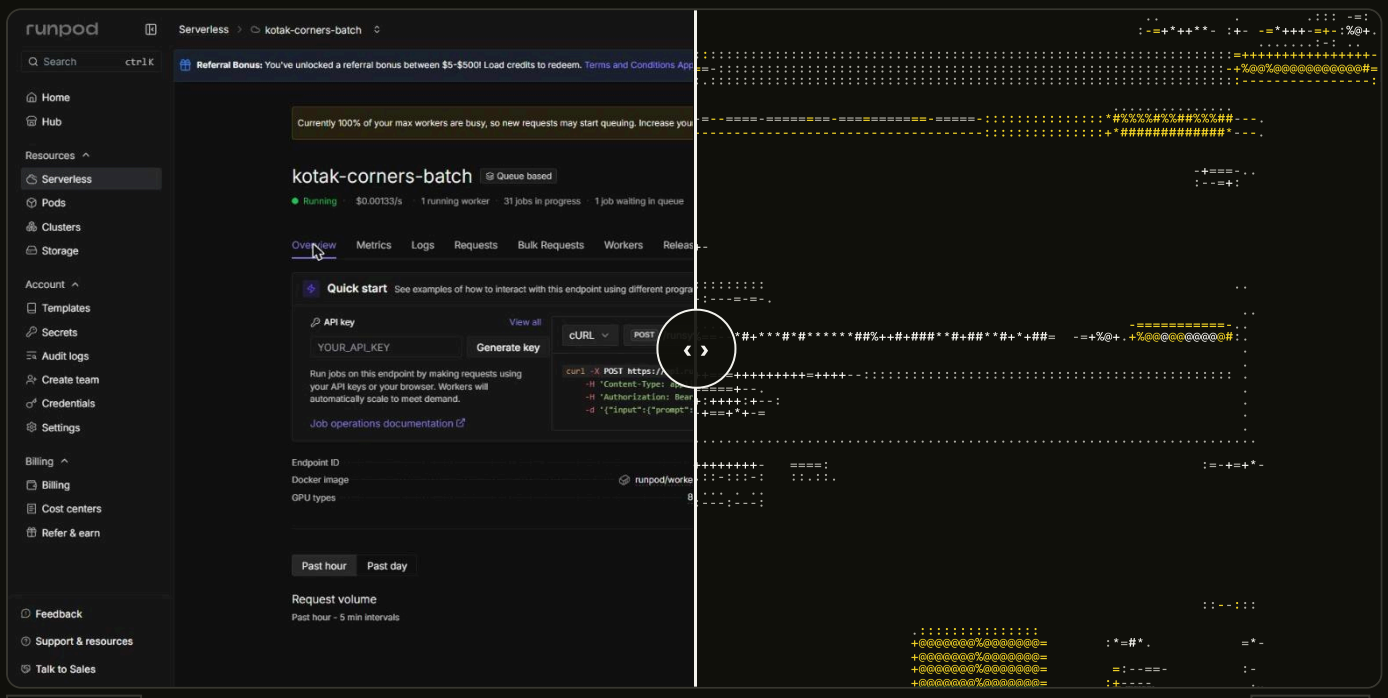
Handbook 10.7 asks for a RunPod dashboard screenshot showing the endpoint URL, the GPU type and the runtime, from the team's own account. The following pages show the serverless endpoint kotak-corners-batch (**https://api.runpod.ai/v2/5yjk19uqmrpbp7**) on one NVIDIA H100 80GB GPU, and the batch runtime. The same file is **submission/SDC2026_T010NG_Runpod_POU.pdf** in the repository.

Proof of Use 22 Oct 2026 HTTP errors **0** of 930

# RunPod

One serverless H100 run drafted the skill corners for **all 923 O*NET occupations** Kotak checks people against, ahead of time and once.

- kotak-corners-batch
- Qwen3.8-27B-FP8
- 1 × H100 SXM 80 GB




CONSOLE Fig 1, the endpoint overview — the same screen, redrawn in glyphs GLYPHS


01 What ran on RunPod p. 2
02 Handbook checklist p. 2
03 Workload p. 3
04 Example request p. 3
05 Console figures 1–5 p. 4
06 Sources p. 6


| 923 | 1,285 s | 2.23 M | 0 |
|---|---|---|---|
| occupations drafted | GPU runtime, smoke + full run | tokens in and out of Qwen3.8-27B | roles failed; 7 answers redrawn |

# 01 What ran on RunPod

Source: RunPod console and data/generated/role-corners-run.json

| | | |
|---|---|---|
| 01 | **RunPod account** | Team account (AlaskanTuna), funded by the SDC RunPod voucher: US$25 credit redeemed 22 Oct 2026 |
| 02 | **Product** | Serverless, queue-based endpoint `kotak-corners-batch` |
| 03 | **Endpoint ID** | `5yjk19uqmrpbp7` |
| 04 | **Endpoint URL** | `https://api.runpod.ai/v2/5yjk19uqmrpbp7` (OpenAI-compatible route `/openai/v1`) |
| 05 | **Worker image** | `runpod/worker-v1-vllm:v2.28.0` (vLLM v0.30.0, CUDA 13.0), the stock RunPod vLLM worker |
| 06 | **Model** | `Qwen/Qwen3.8-27B-FP8`, open weights under Apache 2.0 |
| 07 | **GPU** | 1 × NVIDIA H100 80GB HBM3 (H100 SXM), data centre AP-IN-1; pools 80 GB PRO and 80 GB; max 1 worker |
| 08 | **Runtime** | 1,285 s in total: smoke run 502 s (with cold start), full run 783 s |
| 09 | **Volume** | 930 requests, 0 HTTP errors, 7 answers rejected by the validator and regenerated, 0 roles failed |
| 10 | **Tokens** | 1,092,522 prompt, 1,137,450 completion |
| 11 | **Cost** | About US$1.25: account balance US$25.00 before, US$23.75 after (RunPod console, Billing page, 22 Oct 2026 16:05 MYT) |

# 02 Handbook checklist

Participant Handbook sections 10.7 and 9.4

**[x] Endpoint URL, in place of a Pod ID** — Fig 1 [1] [2]

10.7: a Serverless team shows its endpoint URL. The endpoint ID and the `api.runpod.ai/v2/5yjk19uqmrpbp7` URL both appear on the endpoint page.

**[x] GPU type** — Fig 1 [1] · Fig 2 [2]

The endpoint's GPU pools (80 GB Pro, 80 GB), and the H100 SXM the worker actually ran on.

**[x] Runtime** — Fig 2 [3] · Fig 4 [2]

The worker's uptime mid-run (11 m 16 s) and the endpoint's execution time. The run record gives the exact total: 1,285 s.

**[x] The team's own account** — Fig 1 [5] · all figures

Every figure is the same signed-in console, with the account's balance and avatar top right.

**[x] GPU work runs on RunPod** — Section 03, Workload

9.4: this batch is Kotak's only GPU work. No model is fine-tuned or self-trained, and the live app calls the Claude and ElevenLabs APIs instead of running a model.

03

# Workload

Code: scripts/corners/run-batch.ts
Notes: runpod/README.md

Kotak turns any job into six skill corners and asks twelve yes/no questions about them. RunPod generates that content once, ahead of time, for every one of the 923 O*NET occupations. There is no custom handler: the stock RunPod vLLM worker serves the model. This is Kotak's only GPU work: the live app calls the Claude and ElevenLabs APIs (Technical Declaration, sections 3 and 4).




| | | |
|---|---|---|
| 01 | **Input** | The job's title, description and O*NET tasks, grouped by work activity into at most six groups by fixed rules, with no AI. |
| 02 | **Ask** | The model groups the tasks into six named corners and writes three can-do statements per corner (basic, working, advanced), with learning hours, SDG targets and search-link resources. |
| 03 | **Constrain** | A strict JSON schema goes in `response_format`; vLLM's structured outputs make every token follow it. |
| 04 | **Check** | A validator rejects any answer that drops a task, puts a task in two corners or cites a task the job does not have. |
| 05 | **Retry** | A rejected answer goes back to the model with its errors. Seven were regenerated; none failed. |
| 06 | **Write** | `data/generated/role-corners.json` feeds the check-up questions, the Future Skills Radar and the roadmap; `role-corners-run.json` records every run. |

04

# Example request

Endpoint settings: runpod/README.md

request.sh — POST /openai/v1/chat/completions

```
curl https://api.runpod.ai/v2/5yjk19uqmrpbp7/openai/v1/chat/completions \
  -H "Authorization: Bearer $RUNPOD_API_KEY" \
  -H "Content-Type: application/json" \
  -d '{
    "model": "Qwen/Qwen3.8-27B-FP8",
    "messages": [{"role": "user", "content": "Name one skill of a graphic designer as JSON."}],
    "temperature": 0,
    "max_tokens": 50,
    "chat_template_kwargs": {"enable_thinking": false},
    "response_format": {"type": "json_schema", "json_schema": {"name": "skill", "schema": {
      "type": "object", "properties": {"skill": {"type": "string"}},
      "required": ["skill"], "additionalProperties": false}}}
  }'
```

rerun.sh — RUNPOD_API_KEY and RUNPOD_ENDPOINT_ID set

```
# smoke run on the three demo roles, then the full run; it skips jobs already done
bun run batch:corners --codes 15-1254.00,15-1255.00,43-3031.00
bun run batch:corners
```

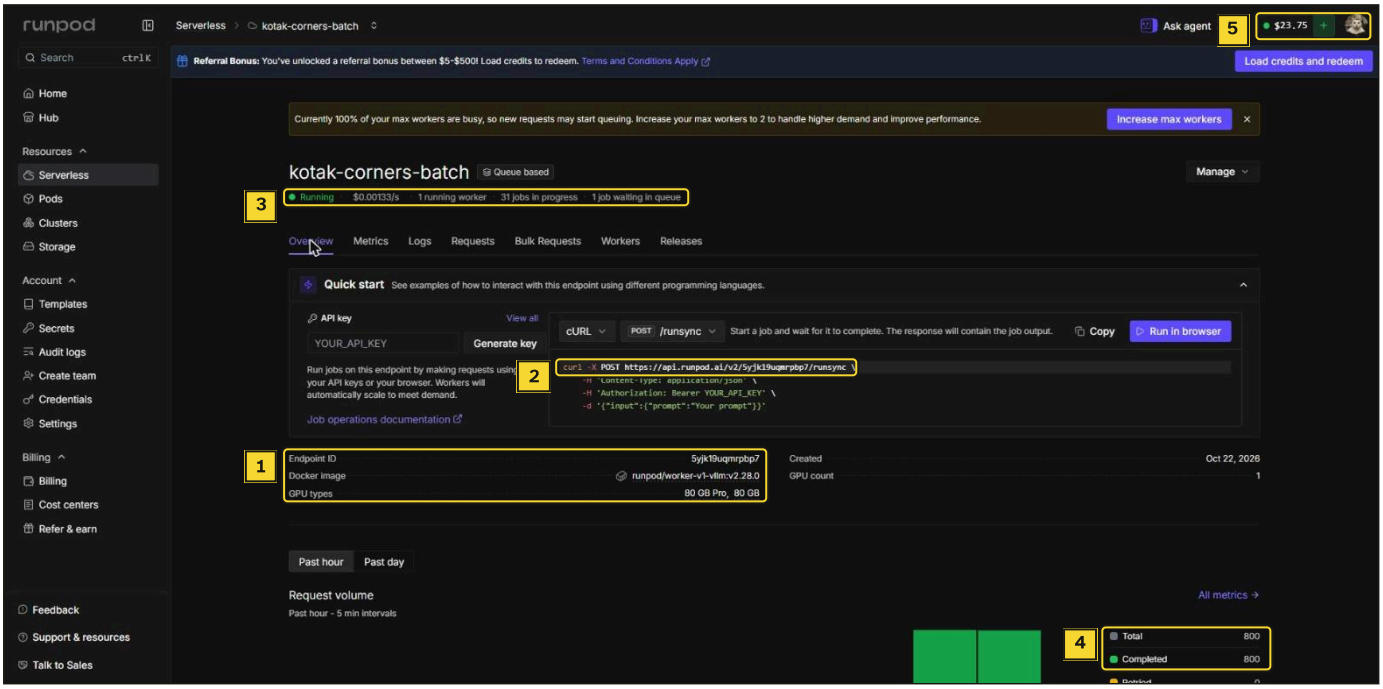
05

# Console figures

Captured during the full run,
22 Oct 2026, about 16:00 MYT

## FIG 1 Endpoint overview

16:00 MYT




1 Endpoint ID 5yjk19uqmrpbp7, the worker image and the GPU pools: 80 GB Pro, 80 GB.

2 The endpoint URL, in RunPod's own cURL example.

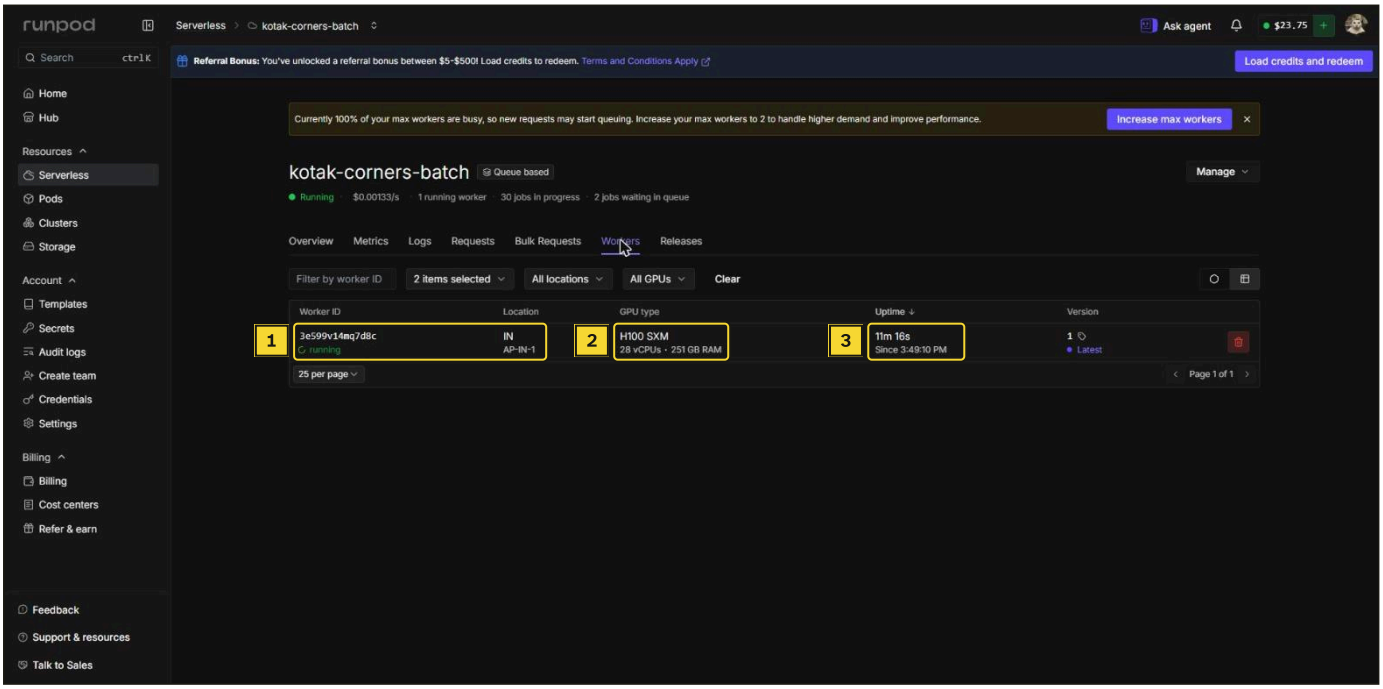
3 Running, with 1 worker and 31 jobs in progress.

4 800 requests completed in the past hour.

5 The account's balance, US$23.75, and avatar.

## FIG 2 Workers tab

during the full run



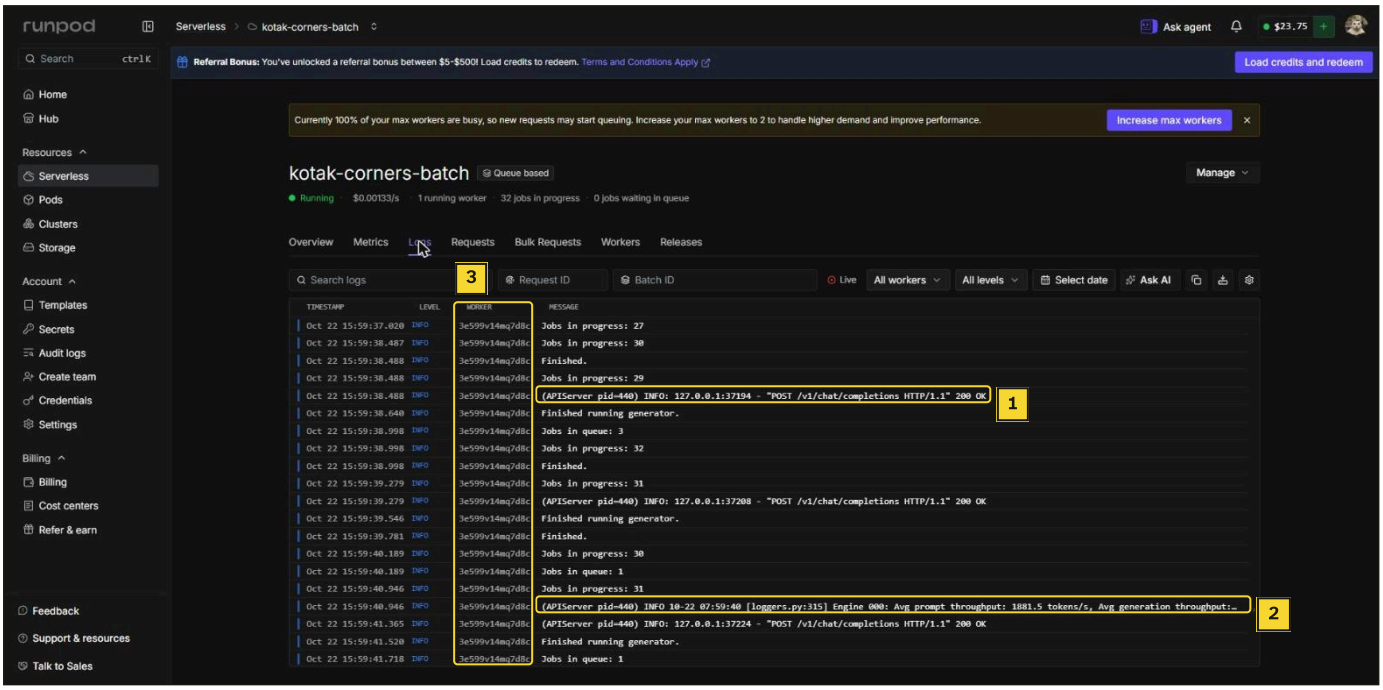

1 Worker 3e599v14mq7d8c, running in AP-IN-1.

2 GPU type: H100 SXM, 28 vCPUs, 251 GB RAM.

3 Uptime 11 m 16 s, since 3:49:10 PM.

## FIG 3 Logs tab

15:59 MYT




1 vLLM serving `POST /v1/chat/completions` with 200 OK.

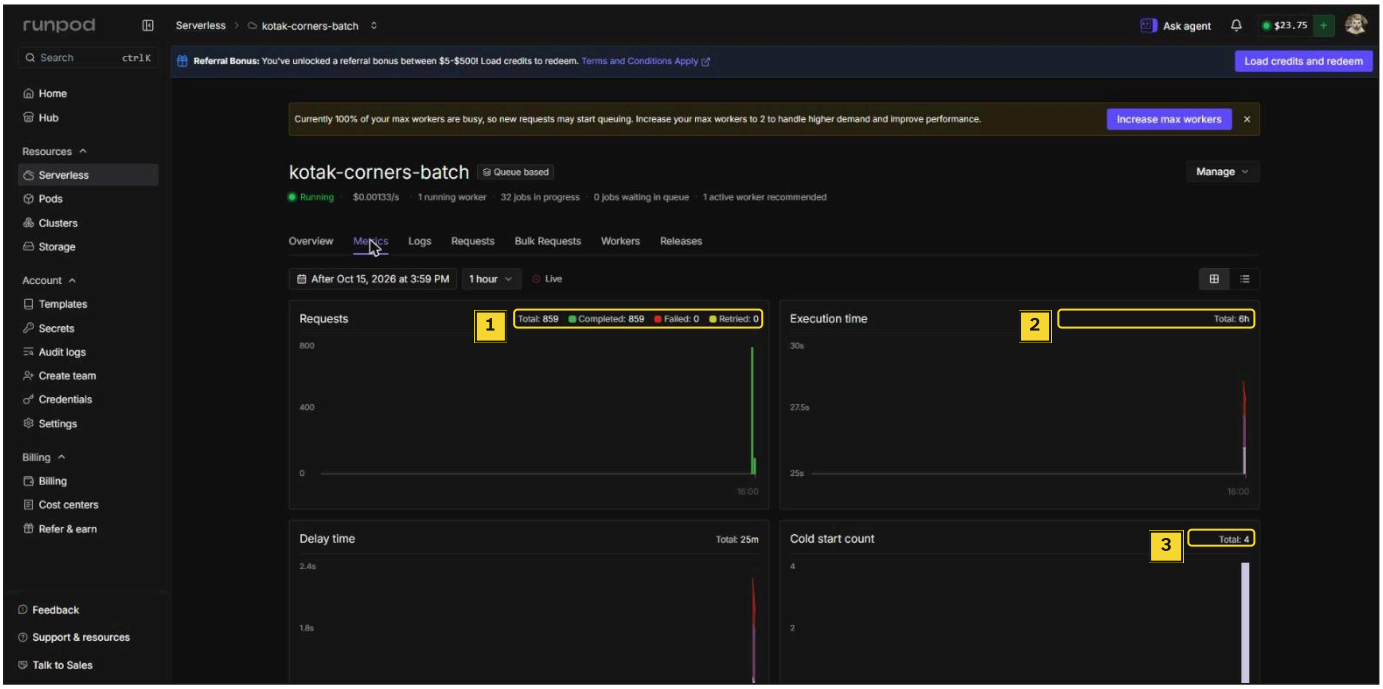
2 The engine's average prompt throughput, 1,881.5 tokens/s.

3 Every line comes from worker 3e599v14mq7d8c.

About 30 jobs are in progress at once: the runner sends 32 requests in parallel.

## FIG 4 Metrics tab

at capture time



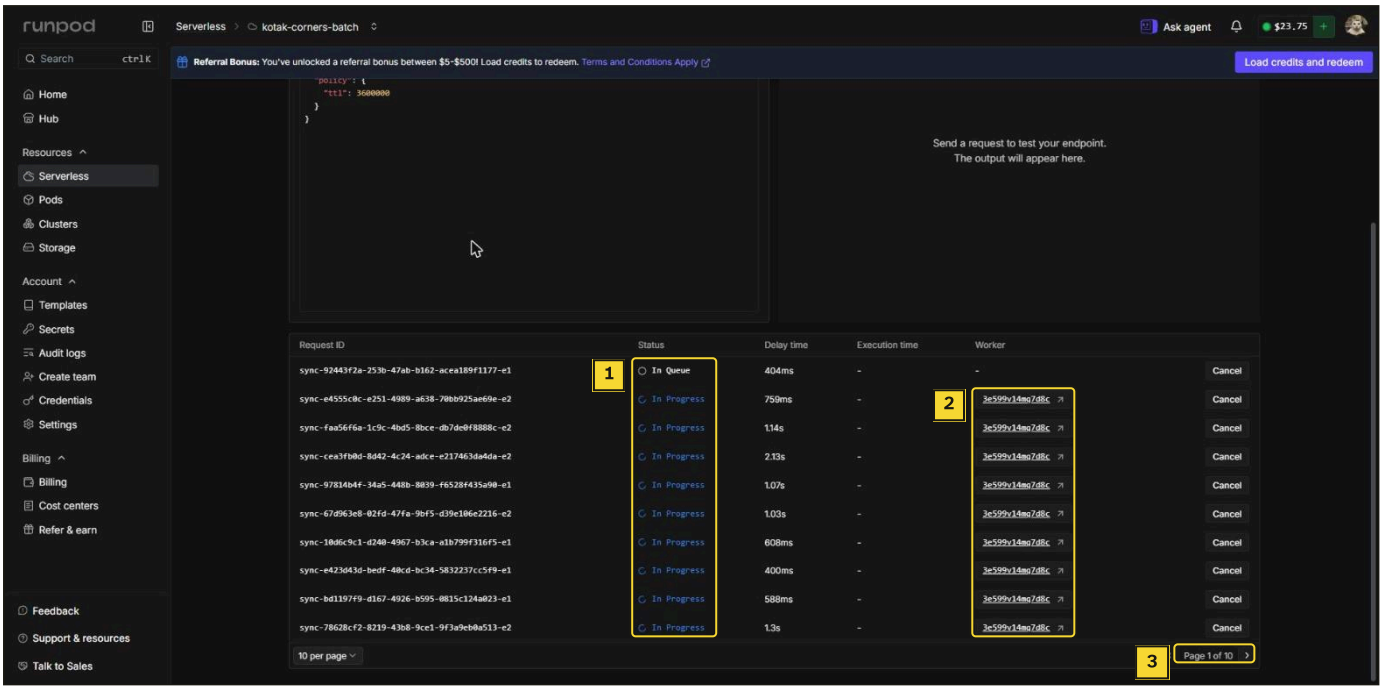

1 859 requests completed, 0 failed, 0 retried.

2 Execution time, 6 h summed over parallel jobs.

3 4 cold starts across both runs.

The filter starts on 15 Oct, so the totals cover both runs; the bars sit at 16:00 on 22 Oct.

**FIG 5** **Requests tab** during the full run




1 Batch jobs in progress and in queue.

2 Each one on worker 3e599v14mq7d8c.

3 Page 1 of 10 of the request list.

06 · Sources

# Where every number here comes from.

| | |
|---|---|
| Console | Figures 1–5, 22 Oct 2026 |
| Run record | data/generated/role-corners-run.json |
| Deploy notes | runpod/README.md |
| Credit, cost | Handbook 9.1.2; RunPod Billing page |
| This PDF | scripts/proof-of-use/runpod.html |
| Repository | github.com/TolongLabs/Kotak |

